

AI506 – Class 10 – GNNs

Gianluca Barmina – `gbarmina@imada.sdu.dk`

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The given code is an implementation of a Graph Neural Network. Consider the code and do the following exercises.

Exercise 1

Read and understand the code and pipeline for this experiment with GNNs. In particular look at e.g. data, inputs, shapes and how the graph is represented.

Exercise 2

Fill in the missing parts of the code (marked as TODOs):

- GNNLayer's forward pass with the aggregate and update steps. Pay attention to the weights matrix shape.
- Add self loops function (needed for taking into account the features of nodes in the diagonal of the adjacency matrix).
- Normalization of the adjacency matrix by: calculating the degree vector (degree of each node), inverting and doing the square root of the degree vector (square root could be inf for 0 degree nodes, set those to 0), creating the diagonal degree matrix, performing and returning the actual normalization.

Run the code. Loss should be very low and accuracy 100%.